28/04/26

SQL Fundamentals: select and Filtering

Question 1

```
SELECT (*)
FROM employees
```

id	first-name	last-name	department	salary	expire-date	city
1	Alice	Green	IT	70 000	2020-01-01	Johannesburg
2	Brian	Lee	HR	45 000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65 000	2018-07-18	Durban
4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

Question 2

```
SELECT DISTINCT department
FROM employees;
```

Department
IT
HR
Finance
Marketing

Question 3

```
SELECT first name,
       Last name
FROM employees
ORDERBY salary DESC;
```

First name	Last name
Eve	Naidoo
Alic	GREEN
Cathy	Zulu
David	Mokwena
Brain	Lee

Question 4

```
SELECT id,
       first name,
```

last name,
salary

FROM employees
ORDER BY salary Desc

id	First name	last name	Salary
5	Eva	Naidoo	72000
1	Alice	Green	70000
3	Cathy	Zulu	65000

Question 5

```
SELECT id,
       first name,
       last name,
       department

FROM employees
WHERE department = 'IT';
```

id	first name	last name	Department
5	Eva	Naidoo	IT
1	Alice	GREEN	IT

Question 6

```
SELECT id,
       first_name,
       last_name,
       department,
       salary
FROM employees
WHERE department = 'Finance' AND salary > 6000;
```

id	first name	Lastname	department	salary
3	Cathy	Zulu	Finance	65000

Question 7

```
SELECT id,
       first_name,
       last_name,
       department

FROM employees
WHERE department = 'HR' OR 'Marketing';
```

id	First_name	Last_name	department
2	Brian	Lee	HR
4	David	Mokwena	Marketing

Question 8

```
SELECT id,
       first_name,
       last_name,
       department

FROM employees
WHERE department NOT IN 'IT';
```

Id	First_name	Last_name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

Question 9

```
SELECT id,
       first_name,
       last_name,
       department

FROM employees
WHERE department = 'IT' IN 'HR' IN 'Finance';
```

id	First_name	Last_name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	marketing
5	Eva	Naidoo	IT

Question 10

```
SELECT id,
       first_name,
       last_name,
       department,
       salary,
       city

FROM employees
~~WHERE department = salary > 65000 AND~~
WHERE department = 'IT' AND salary > 65 000 AND
city = 'Johannesburg';
```

id	First_name	last_name	department	salary	City
5	Eva	Naidoo	IT	72 000	Johannesburg
1	Alice	Green	IT	70 000	Johannesburg